

What is Grid Model in CSS?

-> CSS Grid Layout is a 2-dimensional layout system used to create web page layouts using rows and columns.

-> Grid helps developers arrange items in both horizontal and vertical directions easily.

Why Grid is Used?

-> Before Grid, developers used:

- 1.Floats
- 2.Position
- 3.Inline-block
- 4.Flexbox

-> But creating complex layouts was difficult.

-> CSS Grid solves this problem.

Grid Works in 2 Directions:

Horizontal → Columns

Vertical → Rows

Difference Between Grid and Flexbox:

CSS Grid	Flexbox	
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2D Layout System	1D Layout System	
Works with rows + columns	Works in one direction only	
Best for page layouts	Best for small component layouts	
More control over layout	Better for alignment	

| Complex layouts | Simple layouts |

What is Grid Container and Grid Items?

1. Grid Container: The parent element with `display:grid` is called Grid Container.
2. Grid Items: All direct child elements inside grid container are called grid-items.

What are Grid Column Lines and Grid Row Lines?

1. grid-column-lines: The lines between column is called as grid-column-lines.
2. grid-row-lines: The lines between row is called as grid-row-lines.

Important Concept: (n+1) Here, n = no. of columns

If there are 3 columns = There will be 4 column lines

Grid Parent Properties:

1. `display:grid` = Converts normal container into grid container.
2. `grid-template-columns` = It is used to define number and size of columns.

values: `grid-template-columns: 100px 200px 300px;`

`10% 20% 30%;`

`1fr 2fr 3fr;`

`auto auto auto;`

`repeat(3, 100px)` or `repeat(3, auto)`, or `repeat(3, 1fr)`

`100px 1fr auto;` // combination also works

3. grid-template-rows = It is used to define number and size of rows.

values: grid-template-rows: 100px 200px 300px;

10% 20% 30%;

1fr 2fr 3fr;

auto auto auto;

repeat(3, 100px) or repeat(3, auto), or repeat(3, 1fr)

100px 1fr auto; // combination also works

4. gap: Adds spacing between rows and columns.

Syntax: gap: row-gap column-gap;

5. justify-items: Aligns items horizontally inside each grid cell.

Syntax: justify-items: stretch (default=Full width) | start | center | end;

6. align-items: Aligns items vertically inside each grid cell.

Syntax: align-items: stretch (default=Full height) | start | center | end;

Grid Child Properties:

1. grid-column-start and grid-column-end:

-> It is used to merge the one or more columns (horizontally).

-> Shorthand Value: grid-column: grid-column-start / grid-column-end;

-> Example: grid-column: 1/3;

2. grid-row-start and grid-row-end:

-> It is used to merge the one or more rows (vertically).

-> Shorthand Value: `grid-row: grid-row-start / grid-row-end;`

-> Example: `grid-row: 1/3;`